

WASP-12b

R-0306

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_240204 · created 2026-07-17T21:40:30+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460344.7979 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1462 +/- 0.0099
Transit depth (Rp/Rs)^2	2.14 +/- 0.29 [%]
Orbital Inclination (inc)	82.5 +/- 2.78
Airmass coefficient 1 (a1)	1.021 +/- 0.014
Airmass coefficient 2 (a2)	-0.0159 +/- 0.0025
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#2 - [446, 299]
Optimal Aperture	4.76
Optimal Annulus	12.85
Transit Duration (day)	0.1257 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1462 $\pm$ 0.0099 vs 0.1178 (deviation 2.22 σ)
Duration fit vs published	0.1257 d vs 0.125 d (deviation 0.08 σ)
Mid-time O–C	2.0 min
Depth SNR	11.4
Residual scatter	1.35 %

Light curve

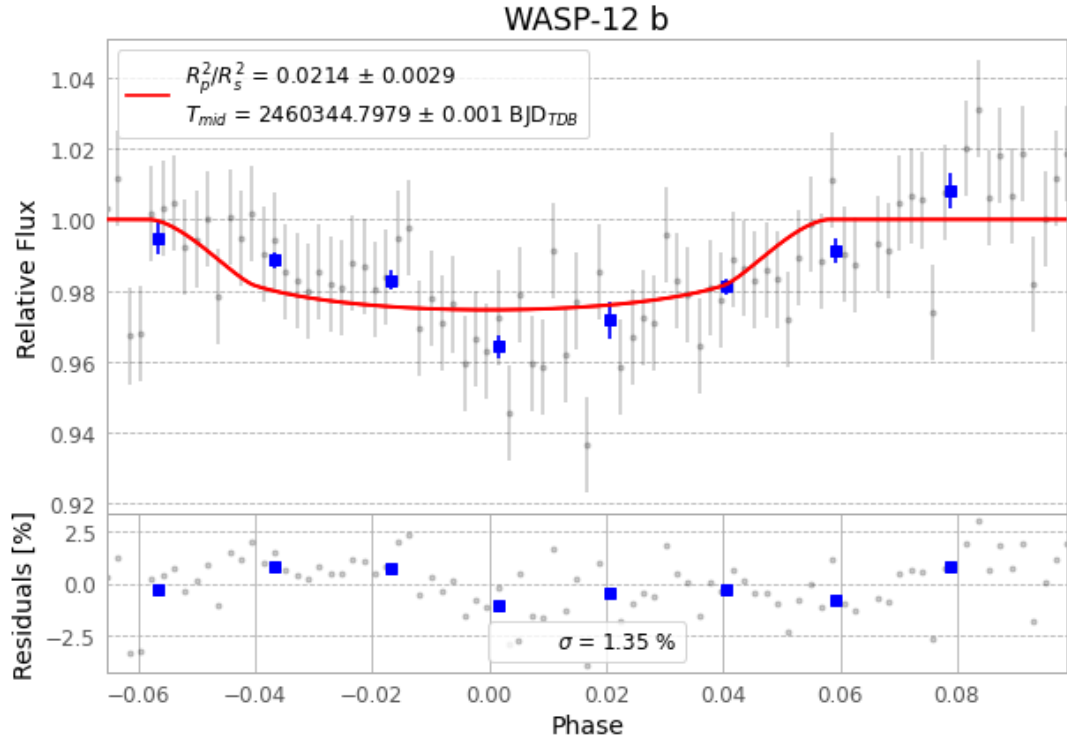


Figure 1 — FinalLightCurve_WASP-12 b_2024-02-03.png (SHA-256 5d8a0e4d70fa54bb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fd295ef180744b15...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 055cf2e

Data provenance

87 FITS frames (57 MB), WASP-12240204051815.FITS → WASP-12240204093615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-240204.log (668a8b4e287d...), FinalLightCurve_WASP-12 b_2024-02-03.png (5d8a0e4d70fa...), FinalLightCurve_WASP-12 b_2024-02-03.csv (81a89fea7215...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 21:40Z.